

Entropy: Process Discovery Experiment Report

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1 Introduction

This report summarizes the setup and results of a process discovery experiment conducted using the *Entropy* tool. The following section outlines the discovery techniques used and their configurations in the experiment. Section 3 specifies the evaluation criteria used to assess the quality of the discovered models and to compare the performance of the discovery techniques. Finally, Section 4 presents the experimental results.

2 Discovery

The command line used to execute the experiment is listed below:

```
-mspd -el "BPI2012.xes" m1 -p 10 -mxItr 20 -t 5000 -mms 500 -pfs 200 -ebrm  
U - optm DFFA
```

Table 1 summarizes the key characteristics of the experimental setup used for performance comparison.

Table 1: Characteristics of the experiment

Parameter	Value
Population size	10
Iterations	20
Time limit (s)	5000
Pareto front size	200
Maximum model size	500
DCI grid siz	4
Optimized models	DFFA
Entropic relevance background model	U
Event log	BPI2012.xes

These discovery techniques are used in the experiment:

- DE [4];

3 Quality

The quality of the discovered models is assessed using two criteria: *entropic relevance*, which captures stochastic precision and recall [2], and *model size*, defined as the number of nodes and edges, which reflects model simplicity.

The performance of the discovery algorithms is evaluated using two aggregate measures: *dominance count* and the *Diversity Comparison Indicator* (DCI) [3]. Dominance count quantifies how many models discovered by a given algorithm dominate those from other algorithms in the global Pareto front. DCI measures the diversity of the discovered models by analyzing their distribution in the objective space. The details are provided in [1];

4 Results

Figure 1 presents the dominant discovered models with respect to entropic relevance and model size, where lower values are preferred for both criteria. Specifically, Fig. 1a, Fig. 1b, and Fig. 1c show the discovered deterministic frequency finite automata (DFFAs), stochastic deterministic finite automata (SDAGs), and directly-follows graphs (DFGs), respectively. The SDAG models are directly optimized, while the DFFA and DFG models are derived through translations of the optimized SDAGs. Table 5 in Appendix A lists the characteristics of all the discovered models.

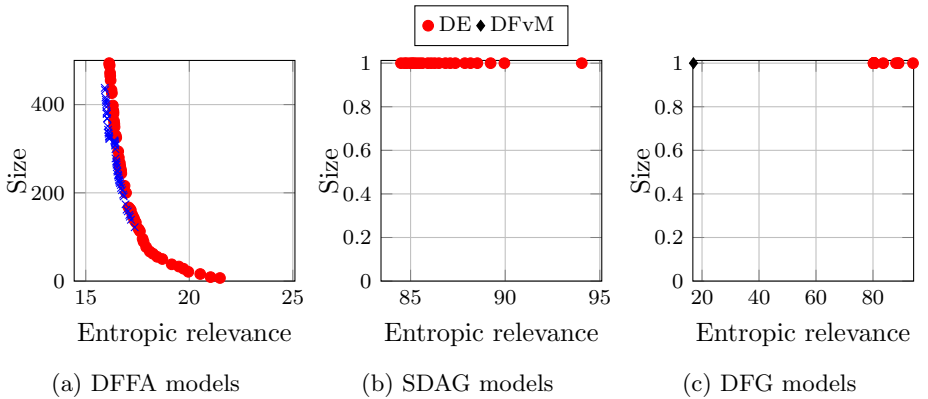


Figure 1: Discovered models plotted by size and entropic relevance for each representation type.

Table 2 and Table 3 summarize the analysis of the diversity of the models and the global Pareto front (dominance count), respectively. Table 4 reports the execution time (in iterations per second) for each input event log and method.

Table 2: Pareto dominance count comparison

Event log	DE(%)	Total
BPI2012.xes	100.00	5 100.00%

Table 3: DCI comparison (%)

Event log	DE(%)
BPI2012.xes	50.00

Table 4: Execution time comparison (iterations per second)

Event log	DE
BPI2012.xes	0.14

References

- [1] Hootan Zhian, Rajkumar Buyya, and Artem Polyvyanyy: *Multi-Objective Metaheuristics for Effective and Efficient Stochastic Process Discovery*. Proceedings of the 23rd International Conference on Business Process Management (BPM) Seville, Spain, August 31–September 5, 2025.
- [2] Hanan Alkhamash, Artem Polyvyanyy, Alistair Moffat, Luciano García-Bañuelos: *Entropic Relevance: A Mechanism for Measuring Stochastic Process Models Discovered From Event Data*. Inf. Syst. 107: 101922 (2022)
- [3] Miqing Li, Shengxiang Yang, Xiaohui Liu: *Diversity Comparison of Pareto Front Approximations in Many-Objective Optimization*. IEEE Trans. Cybern. 44(12): 2568–2584 (2014) Sander J.J. Leemans, Erik Poppe, Moe T. Wynn: *Directly Follows-Based Process Mining: Exploration & a Case Study*. In: ICPM, pp. 25–32 (2019)
- [4] Rainer Martin Storn, Kenneth Price: *Differential evolution - A simple and efficient adaptive scheme for global optimization over continuous spaces*. Tech. Rep. TR-95-012, International Computer Science Institute, 1947 Center Street, Berkeley (1995)

A Model Characteristics

Table 5 summarizes the quality characteristics of discovered models produced by the selected algorithms.

Table 5: Model characteristics: size and entropic relevance

Method	Type	Name	Visualization	Size	Entropic relevance
DE	DFFA	Candidate1	x	21.0	89.230
DE	DFFA	Candidate2	x	32.0	88.529
DE	DFFA	Candidate3	x	14.0	89.956
DE	DFFA	Candidate4	x	61.0	86.842
DE	DFFA	Candidate5	x	85.0	85.914
DE	DFFA	Candidate6	x	92.0	85.627
DE	DFFA	Candidate7	x	105.0	85.172
DE	DFFA	Candidate8	x	127.0	84.976
DE	DFFA	Candidate9	x	137.0	84.684
DE	DFFA	Candidate10	x	41.0	88.167
DE	DFFA	Candidate11	x	95.0	85.472
DE	DFFA	Candidate12	x	51.0	87.363
DE	DFFA	Candidate13	x	116.0	85.095
DE	DFFA	Candidate14	x	68.0	86.498
DE	DFFA	Candidate15	x	7.0	94.046
DE	DFFA	Candidate16	x	58.0	87.093
DE	DFFA	Candidate17	x	44.0	87.867
DE	DFFA	Candidate18	x	75.0	86.275
DE	DFFA	Candidate19	x	82.0	86.089
DE	DFFA	Candidate20	x	144.0	84.498
DE	DFFA	Candidate21	x	102.0	85.310
DE	DFFA	Candidate22	x	134.0	84.798
DE	SDAG	Candidate1	x	1.0	89.230
DE	SDAG	Candidate2	x	1.0	88.529
DE	SDAG	Candidate3	x	1.0	89.956
DE	SDAG	Candidate4	x	1.0	86.841
DE	SDAG	Candidate5	x	1.0	85.913
DE	SDAG	Candidate6	x	1.0	85.626
DE	SDAG	Candidate7	x	1.0	85.169
DE	SDAG	Candidate8	x	1.0	84.973
DE	SDAG	Candidate9	x	1.0	84.680
DE	SDAG	Candidate10	x	1.0	88.167
DE	SDAG	Candidate11	x	1.0	85.471
DE	SDAG	Candidate12	x	1.0	87.363
DE	SDAG	Candidate13	x	1.0	85.091
DE	SDAG	Candidate14	x	1.0	86.498
DE	SDAG	Candidate15	x	1.0	94.046
DE	SDAG	Candidate16	x	1.0	87.093
DE	SDAG	Candidate17	x	1.0	87.867
DE	SDAG	Candidate18	x	1.0	86.273
DE	SDAG	Candidate19	x	1.0	86.088
DE	SDAG	Candidate20	x	1.0	84.493
DE	SDAG	Candidate21	x	1.0	85.307

Method	Type	Name	Visualization	Size	Entropic relevance
DE	SDAG	Candidate22	x	1.0	84.794
DE	DFG	Candidate1	x	1.0	88.936
DE	DFG	Candidate2	x	1.0	88.079
DE	DFG	Candidate3	x	1.0	88.936
DE	DFG	Candidate4	x	1.0	80.384
DE	DFG	Candidate5	x	1.0	80.384
DE	DFG	Candidate6	x	1.0	80.384
DE	DFG	Candidate7	x	1.0	80.384
DE	DFG	Candidate8	x	1.0	80.242
DE	DFG	Candidate9	x	1.0	80.242
DE	DFG	Candidate10	x	1.0	83.582
DE	DFG	Candidate11	x	1.0	80.384
DE	DFG	Candidate12	x	1.0	80.384
DE	DFG	Candidate13	x	1.0	80.242
DE	DFG	Candidate14	x	1.0	80.384
DE	DFG	Candidate15	x	1.0	94.046
DE	DFG	Candidate16	x	1.0	80.384
DE	DFG	Candidate17	x	1.0	80.384
DE	DFG	Candidate18	x	1.0	80.384
DE	DFG	Candidate19	x	1.0	80.384
DE	DFG	Candidate20	x	1.0	80.242
DE	DFG	Candidate21	x	1.0	80.384
DE	DFG	Candidate22	x	1.0	80.242